

LESSON GUIDE

Technology Plus

Unit 5: Software

3.3.2 – Collaboration Software

Lesson Overview:

In this lesson, students will explore collaboration software and identify how it impacts working with a team.

Students will:

- Explain the purpose of collaboration software.
- Differentiate between the types of collaboration software based on their use.
- Analyze how collaboration software impacts communication, organization, and efficiency compared to manual methods.

Guiding Question: What is collaboration software and how does it impact working with a team?

Suggested Grade Levels: 8-12

Technology Needed: Yes

Approximate Time Required: 90-115 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 3.3 – Explain the purpose and proper use of software.
 - **Collaboration software**
 - **Email client**
 - **Conferencing**
 - **Online workspace**
 - **Document sharing**
 - **Instant messaging software**
 - **Web-browsing software**
 - **Remote support software**

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Lesson Background

Background Information:

Collaboration software refers to digital tools that allow individuals or groups to work together efficiently, regardless of location. These tools can include email clients, conferencing software, online workspaces, document sharing platforms, instant messaging applications, web browsers, and remote support software. Each tool has a specific purpose but shares the common goal of enhancing communication, organization, and productivity. Understanding how to properly use these tools is essential for students, not just for classroom projects, but also for future careers and everyday life.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 3.3.2 – Collaboration Software • Activity Slides 3.3.2 – Paper vs. Pixels
Materials per Student	• Student Notes 3.3.2 – Collaboration Software • Student Handout 3.3.2 – Collaboration Software • Assessment 3.3.2 – Collaboration Software (optional) • Paper • Pencil • Collaboration Software

Lesson Background

Cyber Terms:

Term	Definition
collaboration software	any digital tool designed to help people work together effectively, whether they are in the same location or spread across different places
email client	allows users to send, receive, organize, and manage email communications
conferencing software	used to conduct virtual meetings and discussions, enabling real-time interaction regardless of a participant's location
online workspaces	allows teams to collaborate on the same file simultaneously, assign tasks, track changes, and monitor progress
document sharing	allows users to share files and documents with others, enabling collaborative editing and review
instant messaging (IM) software	provides rapid and convenient means of communication, allowing users to send and receive messages instantly
web-browsing software	an application that typically comes pre-installed and allows users to access and navigate the internet, known as a web browser
remote support software	enables technical support teams or IT (Information Technology) professionals to access and troubleshoot a user's computer or device remotely to resolve issues or perform maintenance tasks

3.3.2 – Collaboration Software

Guiding Question: What is collaboration software and how does it impact working with a team?

Lesson Launch (5-10 minutes)

- **Ask:**
 - What if you had to write and mail a letter every time you wanted to contact someone at a distance?
 - What if you had to schedule an in-person meeting for every important conversation?
 - What if you had to pass paper back and forth during a group project?
- Allow think time and let a few students answer.

Student answers should reflect that communication would take longer, group projects would be harder, and it could be difficult for everyone to put forth the same amount of participation.
- **Say:** Without collaboration software, teamwork would be limited, slower, and less organized. It would be harder to communicate, file sharing could get complicated, projects would take longer, and remote learning and online work would be nearly impossible.

Click, Share, Collaborate: Digital Teamwork (30-40 minutes)

- Distribute the Student Handout 3.3.2 – Collaboration Software and have students fill in the answers during the lesson.
- Introduce **collaboration software**, its definition, and benefits.
 - Collaboration software supports sharing ideas, files, tasks, and real-time updates so teams can stay informed, organized, and productive.
 - Collaboration software also helps with productivity.
- Introduce **email client**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - Users can send and receive messages, manage contacts and calendars, assign themselves tasks and deadlines, and set reminders to follow up on things.
 - Usage:
 - Email Management: Organize messages into a folder or categories to keep inbox manageable.
 - Communication: Use formal and informal communication, including sending attachments, links, and updates.
 - Scheduling: Set up email filters, rules, and automated responses to manage email flow effectively.
 - Integration: Sync with calendars and task managers for improved organization and

productivity.

- Introduce **conferencing software**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - Conferencing software mostly involves video calls where users can see each other by using their web cameras during the conference. It comes with many features like virtual backgrounds, screen sharing, recording, messaging, transcripts, and more.
 - Usage:
 - Video: Use for face-to-face virtual meetings, presentations, and discussions for a more personal touch.
 - Audio: Conduct meetings without video with only voice communication.
 - Screen Sharing: Share a screen to present documents, demonstrate software, or collaborate on visual content.
 - Recording: Record meetings for later review or for those who couldn't attend live.
- Introduce **online workspace**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - In online workspaces, multiple people can collaborate on the same file simultaneously, assign tasks, track changes, and monitor progress.
 - Usage:
 - Task Management: Assign, track, and manage tasks and deadlines within the platform.
 - Communication: Use built-in messaging, discussion boards, or chat features for team communication.
 - Integration: Connect with other tools and apps (like calendar, email, and document storage) for a seamless workflow.
 - Customization: Tailor the workspace to fit the team's needs, including setting up different channels or project boards.
- Introduce **document sharing**, its definition, uses, two examples, and a real-world scenario of the software in action and its features.
 - Most online workspaces feature document sharing. When utilizing document sharing, users should consider backups, storage limit, file synchronization and encryption, and security mechanisms.
 - Usage:
 - Access Control: Set permissions to control who can view, edit, or comment on documents.
 - Version Control: Track changes and maintain different versions of documents to avoid confusion and data loss.
 - Real-Time Collaboration: Use tools that support simultaneous editing, so multiple users can work on a document at the same time.
 - Link Sharing: Share documents through links, ensuring that access is restricted to intended recipients.
- Introduce **instant messaging (IM)** software, its definition, benefits, two examples, and a real-world scenario of the software in action and its features.
 - IM software supports one-on-one and group conversations. Text messages, photos, and files can be sent between phones anywhere if they are connected to a cellular network.

- IM supports professional and personal communication.
- Benefits:
 - Real-Time Communication: Immediate responses and instant updates for urgent messages.
 - Group Chats: Team coordination for sharing updates or making quick decisions.
 - File Sharing: Send and receive files, images, and documents directly through the messaging software.
 - Status Updates: Availability indicators and custom statuses to avoid interruptions or provide context.
 - Integration with Other Tools: Sync with productivity apps to streamline workflow.
 - Security and Privacy: Encryption protects conversations, and access control determines who can message you and view your online status.
 - User Experience: Enhance communication with emojis and GIFs, utilize text formatting (bold, italics) to emphasize information.
- When to use IM Software:
 - Immediate responses
 - Quick clarification
 - Casual communication
 - Coordinating tasks
 - Giving quick updates
- Introduce **web-browsing software**, its definition, benefits, popular examples, and a real-world scenario of the software in action and its features.
 - Users can use the software to access information online, interact with online content, and use tools to navigate, manage, and organize web pages.
 - Benefits:
 - Navigating Web Pages: Use search bar, back and forward buttons, and tabs to navigate to new pages, revisit old pages, and switch pages easily.
 - Search Engines: Configure browser for a preferred search engine and use search suggestions.
 - Bookmarks and Favorites: Bookmark frequently visited websites and categorizes bookmarks for organization.
 - Privacy and Security: Set options including browsing without saving history or cookies, installing security extensions, and regularly clearing cache and cookies.
 - Managing Extensions: Install and manage browser extensions to add functionality.
 - Accessibility Features: Adjust zoom, text size, and screen readers to improve readability and accessibility.
 - Updating Software: Keep browser regularly updated and enable automatic updates where possible.
 - Managing Downloads: Download files from the internet and enable safety precautions so the browser won't download anything suspicious.
 - Examples:
 - Google Chrome
 - Microsoft Edge
 - Apple Safari

- Mozilla Firefox
- When to use:
 - Information search
 - Online transactions
 - Social interactions
- Introduce **remote support software**, its definition, benefits, two examples, and a real-world scenario of the software in action and its features.
 - Remote support software provides efficient and timely assistance without requiring the technician to be physically present.
 - Benefits:
 - Connecting Remotely: The technician initiates a remote session where the user is typically required to provide a session code or grant access through the software.
 - Performing Tasks: Technicians view and interact with the user's machine to diagnose problems or perform actions as if they were present.
 - Communication: Use built-in chat features to communicate during the support session to provide explanations or instructions.
 - Security Measures: Ensure the remote session is encrypted, keep logs of remote sessions, and allow the user to end the session at any time.
 - Session Management: Technicians request only the necessary permissions and set session timeouts.
 - Software Features: Technicians use integrated diagnostic tools to run system checks and analyze performance issues, and record sessions for training, troubleshooting, or review purposes.
 - User Experience: Provide clear instructions to users on how to start and end remote support sessions and offer guidance on how to prepare for a support session.
 - When to use:
 - Technical troubleshooting for issues that can be resolved remotely
 - System maintenance for updates
 - Patches or configuration changes
 - User assistance to help with software usage, setup, or training

Activity: Paper vs. Pixels (45-60 minutes)

- Display Activity Slides 3.3.2 – Paper vs Pixels, have students record their answers in the handout.
- For this activity students will work in groups of 3-4.
- Students will need paper, pencils, and collaboration software.
- Collaboration software should be available school-wide so that everyone has access. If your school does not provide collaboration software, students can still do the 'Without Software' sessions of this activity.
- As a group, students will explore planning a presentation with and without collaboration software and compare the differences.
- Plan to provide students with at least 15 minutes for each collaboration tool, this will allow them time for with/without software, and to answer the follow-up questions.

- Remind students as needed, their presentation is about how technology, specifically productivity and collaboration software, impacts education.
- The goal of this activity is for students to experience the benefits of collaboration software, not to produce a polished presentation. The presentation is just a focal point so students have a purpose while exploring each collaboration tool.

Putting It All Together (10 minutes)

- Review what students learned today by discussing the questions below as a class.
- Imagine your group is working on a big school project but members are spread out at home. Which collaboration tools would you choose to keep the project organized and why?

Students should explain their choices (e.g., email for updates, doc sharing for writing, IM for quick check-ins) and justify how those tools solve real problems.

- Think about a time you've had to work with others. What challenges did you face, and how could collaboration software have made that experience easier?

Students reflect on personal experience (lost files, scheduling issues, miscommunication) and tie it back to the tools in the lesson.

- Collaboration software often has overlapping features (for example, both IM and online workspaces allow messaging). How would you decide which tool to use for a specific situation?

Students think critically about choosing the right tool, not just naming what exists.

- How could collaboration software help you outside of school, such as in a job, club, or personal project?

Students might talk about coordinating schedules, sharing files for a hobby, working on a group presentation at work, or staying connected with teammates remotely.